ANALYTICAL PROBLEM SOLVING

ASSESSMENT TOOL - 1

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COURSE : COMPUTER VISION.

1. GREY LEVEL TRANSFORMATION.

a) Purpose.

* Grey level transformation is used to enhance image quality by changing pixel intensities to improve visibility and contrast.

b) Negative Transformation.

Formula : $s = 255 - r$.

INPUT r	OUTPUT s
50	205
100	155
150	105
200	55

INTERPREATATION: Bridge areas become dark and dark areas become bright, producing a negative image.

2. HISTOGRAM EQUALIZATION.

Histogram :

INTENSITY	FREQUENCY.
0	4
1	6
2	10
3	6

Total Pixels = 26.

a) Purpose.

* Histogram equalization improves image contrast by spreading pixel intensities over the available image.

b) CDF.

INTENSITY	FREQUENCY	CDF
0	4	4
1	6	10
2	10	20
3	6	26

c) Equalized Intensities.

INTENSITY	EQUALIZED VALUE.
0	0
1	1
2	2
3	3

$$S = \frac{CDF}{26} \times 3.$$

INTERPRETATION: The contrast becomes more uniform, making image details easier to distinguish.

3. GAUSSIAN SMOOTHING

a) what why Gaussian filtering?

* Gaussian filtering reduces noise smoothly and preserves edges better than averaging filtering.

b) calculation.

Kernel:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

New center value:

$$= \frac{1}{16}[10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10]$$

$$= \frac{360}{16} = 22.5 \approx 23.$$

INTERPRETATION: Noise is reduced and the image becomes smoother.

4. EDGE DETECTION USING SOBEL OPERATOR.

a) Purpose.

* Edge detection identifies object boundaries by detecting sudden intensity changes.

b) Horizontal Sobel.

Mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ -1 & 2 & 1 \end{bmatrix}$$

image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Gradient:

$$= (-10-20-10) + 0 + (30+60+30)$$

$$= 80.$$

$$\boxed{\text{Gradient} = 80}$$

INTERPRETATION: A Strong horizontal edge is detected.

5. CONTRAST STRETCHING

a) Need.

* Contrast stretching expands the intensity range, making the image clearer and improving visibility.

b) Linear Stretching.

Formula:

$$S = \frac{(r-50)}{150-50} \times 255$$

For $r = 50$;

$$S = 0.$$

For $r = 100$;

$$S = \frac{50}{100} \times 255 = 127.5 \approx 128$$

For $r = 150$;

$$S = 255.$$

INPUT	OUTPUT.
50	0
100	128
150	255.

INTERPRETATION: The intensity range is expanded from 50-150 to 0-255 - resulting in a higher-contrast and more detailed image.